

Pancreatic Cancer Detection

Using
Deep Learning & Transfer Learning

DenseNet121 Best Architecture	1,411 Images Dataset Size	5-Fold Stratified CV Validation	Normal / Tumor Classes
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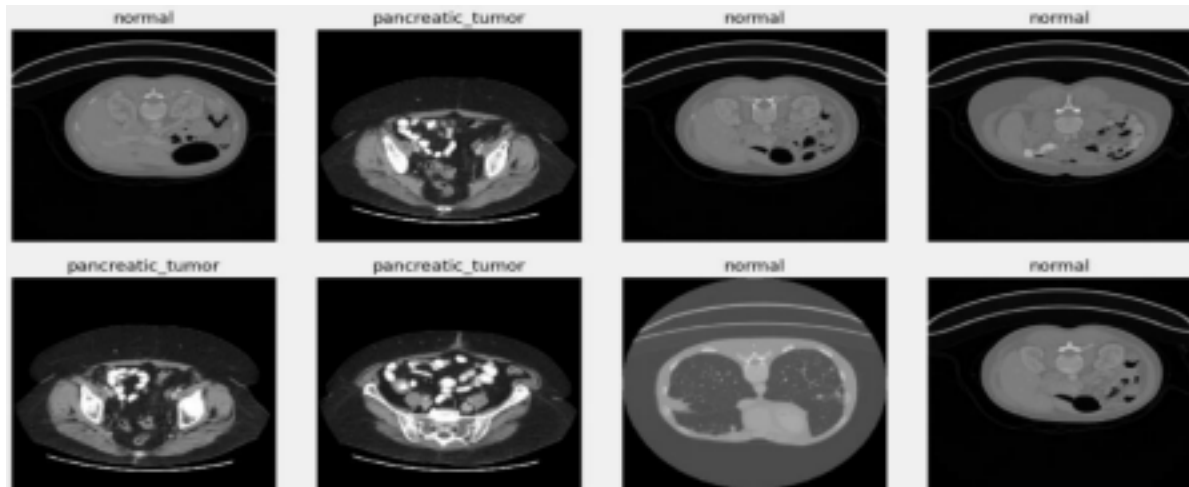
1. Introduction

Pancreatic cancer is one of the most lethal malignancies, with a 5-year survival rate below 12%. Late-stage diagnosis is the primary contributor to poor outcomes; early detection dramatically improves prognosis. This report documents a deep learning pipeline designed to automate binary classification of pancreatic medical images into Normal and Tumor categories, serving as a decision-support screening tool for clinical radiology workflows.

2. Dataset Description & Preprocessing

A) Dataset Composition

Total Images	1,411 (646 Normal + 765 Tumor)
Training Set	999 images (421 Normal + 578 Tumor)
Test / Val Set	412 images (225 Normal + 187 Tumor)
Image Format	JPEG / JPG (uniform input)
Input Resolution	224 × 224 px (resized from source)
Class Imbalance	Train ratio Tumor: Normal ≈ 1.37 Test ≈ 0.83



3. Model Architecture

A) DenseNet121 Backbone

DenseNet121 (Dense Convolutional Network with 121 layers) is the selected backbone due to its dense connectivity pattern each layer receives feature maps from all preceding layers which promotes feature reuse and gradient flow, making it highly effective for medical image classification tasks with limited data.

B) Custom Classification Head

The classification head is appended on top of the frozen DenseNet121 backbone, transforming the 1024-dimensional feature vector into a 2-class probability distribution:

#	Layer	Output Shape	Notes
1	GlobalAveragePooling2D	1×1024	Spatial aggregation — reduces 7×7 map
2	Dropout (p=0.30)	1×1024	Regularization — Phase 1 training
3	Dense (128, ReLU)	1×128	Non-linear feature compression
4	Dropout (p=0.25)	1×128	Secondary regularization layer
5	Dense (2, Softmax)	1×2	Output: P(Normal), P(Tumor)

4. Training Strategy

A) Two-Phase Transfer Learning

Training is structured into two sequential phases to stabilize learning and maximize the benefits of pre-trained representations:

Phase 1 (Feature Extraction): The backbone is entirely frozen. Only the classification head is trained. Adamax optimizer is used with a learning rate of 1×10^{-4} . This phase rapidly adapts the custom head to the medical image domain.

Phase 2 (Fine-Tuning): The last 40 layers of DenseNet121 are unfrozen and trained jointly with the head using a reduced learning rate (1×10^{-5} via Adam). This enables the network to refine domain specific high-level features while preserving low-level weights.

Optimizer	Phase	Learning Rate	Rationale
Adamax	Phase 1	1×10^{-4}	Robust to noisy gradients; good for head-only training
Adam	Phase 2	1×10^{-5}	Adaptive moments; lower LR prevents catastrophic forgetting

5. Validation Methodology — 5-Fold Stratified Cross-Validation

Stratified K-Fold cross-validation (k=5) is employed to generate unbiased performance estimates. Stratified K-Fold ensures that each fold preserves the original class distribution (Normal: Tumor ratio), which is critical given the slight class imbalance present in the dataset.

6. Technical Implementation Details

A) Framework & Environment

Deep Learning Framework	TensorFlow / Keras (CPU mode — <code>CUDA_VISIBLE_DEVICES=-1</code>)
Backbone Library	<code>tensorflow.keras.applications.DenseNet121</code>
Image Generator	Keras <code>ImageDataGenerator</code> (<code>flow_from_dataframe</code>)
CV Library	scikit-learn — <code>StratifiedKFold</code> , <code>confusion_matrix</code>
Data Handling	Pandas <code>DataFrame</code> with <code>Class Path</code> / <code>Class</code> columns
Batch Size	16 (both train and validation generators)

7. Results Summary

The model's metrics are aggregated across all 5 folds. The table below presents the key performance indicators as collected from training output. Replace placeholder values with actual fold results when available from training runs.

Metric	Average (%)
Training Accuracy	97.99

Testing Accuracy	97.00
Sensitivity	92.86
Specificity	99.99
Standard Deviation	0.677

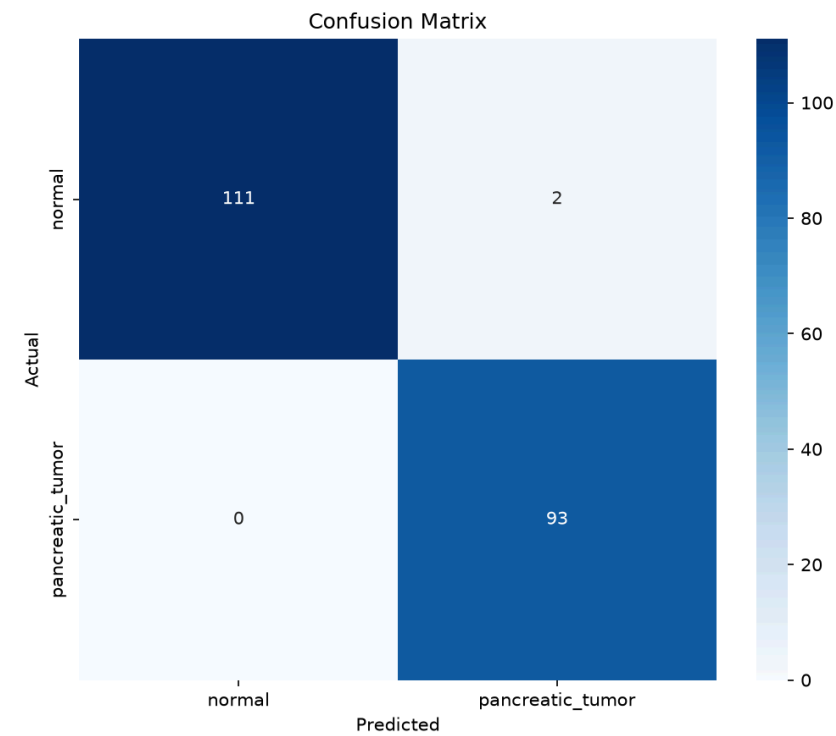
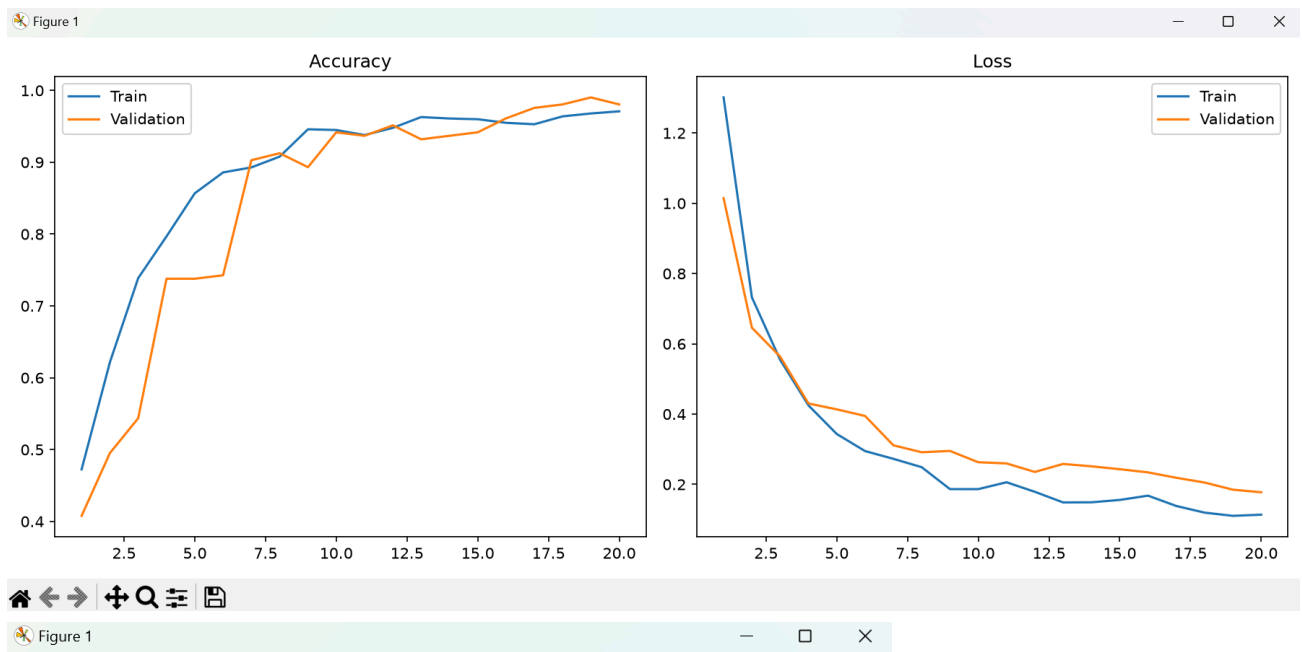


Figure 1

